

AMOS Professional



AMOS Professional - Developer Manual v1.0

For AmosLikeBasic Engine (macOS/Windows)

Welcome to your new AMOS environment. This document covers the commands implemented in your engine up to December 2026.

1. Introduction	3
2.  Input & Conditions (Control)	4
3.  Text & Log (Console)	4
4.  Graphics	6
5.  Sprites	8
6. Tilemap & Levels	8
7. Sound & Music	8
8. Functions	9
9. List of alla commands	10
10. List of alla graphics commands	11
11. List of alla functions	12
12. C64 Palette Index	12

1. Introduktion

AMOS Professional låter dig skapa avancerade grafiska program och spel med ett Basic-liknande språk. Miljön stödjer flera kommandon på samma rad separerade med : och kommentarer med ;.

The environment supports multiple commands on the same line with : and comments with ;

2. Input & Conditions (Control)

REM Text

Example: REM Loop 10 times or just ; Loop 10 times

LET variabel=x

Example: Let X=0 or X=0

DIM variable(x)

Example: DIM A(10)

KEYSTATE("KeyName")

Returns 1 if the key is pressed, otherwise 0.

Keys: "Left", "Right", "Up", "Down", "Space", "A", etc.

Example: IF KEYSTATE("Space") = 1 THEN PRINT "FIRE!"

IF condition ELSE statement THEN statement

Runs a command if the condition is true.

Example: IF X > 640 THEN LET X = 0

IF X > 640 THEN

X=0

ELSE

X=10

END

FOR var = start TO end [STEP s] / NEXT

Classic loop structure.

Example: FOR T = 1 TO 100 : PLOT T, 10 : NEXT T

WHILE / WEND

REPEAT / UNTIL

SELECT / CASE /DEFAULT / ENDSELECT

INC / DEC

DATA / RESTORE / READ

GOTO label / GOSUB label / RETURN

Jumps to a label (e.g., MainLoop:) or runs a subroutine.

3. Text & Log (Console)

PRINT "Text" / PRINT AT row, col, "Text"

PRINT writes text to the text console on the screen.

PRINT AT places text on a specific row and column.

LOCATE row, col

Moves the text cursor for the next PRINT command.

INPUT "Text", variable

Programmer Tips

Always include a WAIT VBL in your main loop to keep graphics smooth.
Use REM to write comments in your code.

4. 🎨 Graphics

SCREEN width, height

Creates a new graphics screen with the specified resolution.

Example: SCREEN 640, 480

SCREEN SELECT x

Creates / selects screen layer x

LOAD "filename.png"

Loads an image as the background image for the entire screen.

Example: LOAD "forest.png"

LOAD x, "filename.png"

Loads an image as BOB number x.

Example: LOAD 1, "house.png"

CLSG x (Clear Graphics)

Clears the graphics layer x screen and fills it with the current background color.

WAIT VBL / WAIT milliseconds

WAIT VBL: Pauses the program until the next screen update (sync).

Provides smooth animation.

WAIT 1000: Pauses the program for 1000 milliseconds (1 second).

INKG ColorName, number or #HEX

Sets the current pen color for drawing routines.

Example: INK Red, INK "Red", INK 1 (C64 Color), INK "#FF00FF"

PAPERG ColorName, number or #HEX

Sets the background color for drawing.

CLSG after paper sets the entire background as color x

PLOT x, y

Draws a single pixel at the specified coordinates.

LINE x1, y1, x2, y2

Draws a line between two points.

BOX x1, y1, x2, y2

Draws the outline of a rectangle.

BAR x1, y1, x2, y2

Draws a filled rectangle.

CIRCLE x1,y1,r1

Draws the outline of a circle.

CIRCLEF x1,y1,r1,r2

Draws an filled circle/ellipse

FILL x1,y1

Flood fill

CONFIGFONT width, height, fontname
Configure font and size
Example: CONFIGFONT 8,16,"Courier New"

TEXT x, y, "String"
Draws text directly on the graphics at the pixel level.
Example: TEXT 100, 100, "HELLO WORLD"

SCROLL layer, x, y
Scrolls the entire background smoothly. Perfect for shoot-em-ups.
Example: SCROLL 1, 2, 0 (Horizontal scroll)

BOB number,x,y,image
Example: BOB 1,100,150,1

5. Sprites

`SPRITE LOAD id, "filename.png"`

Loads an image file as a sprite. The first pixel (0,0) is automatically used as the transparent color.

Example: `SPRITE LOAD 1, "hero.png"`

`SPRITE POS id, x, y`

Places a sprite on the screen.

`SPRITE HANDLE id, offset_x, offset_y`

Sets the sprite's "handle" (hotspot).

Example: `SPRITE HANDLE 1, 16, 16` (Centers the sprite if it is 32x32)

`SPRITE ON id / SPRITE OFF id`

Shows or hides a sprite on the screen.

6. Tilemap & Levels

`TILE LOAD "[file]", width, height`

Loads a "bank" of tiles (graphics blocks) of a given size.

`TILE MAP width, height`

Sets the size of the level in tiles.

`TILE SET x, y, id`

Places a specific tile at coordinates x, y in the level.

`TILE DRAW ox, oy`

Draws the level on the screen with an offset (for scrolling).

7. Sound & Music

`MUSIC PLAY "[file]"`

Plays a tracker module (.mod, .xm, etc).

`MUSIC STOP`

Stops all currently playing music.

`SAM PLAY "[file]"`

Plays a sound effect (sample/wav).

`SAM STOP ("file")`

Stop sound current sound effect or all

8. Functions

INKEY\$

STR\$

CHR\$

ASC

VAL

SIN

COS

RND

INT

HEX

INC

DEC

HIT

TILE

KEYSTATE

LEN

TRIM\$

LOWER\$

LEFT\$

RIGHT\$

MID\$

REPLACE\$

WORD\$

INSTR

9. List of alla commands

CLS
PRINT
INPUT
PAPER
INK
LOCATE

LET
DIM

SELECT / ENDSELECT

END

REM

GOTO / GOSUB / RETURN

IF / THEN / ELSE / ENDIF

CASE / DEFAULT / ENDSELECT

FOR / NEXT

WHILE / WEND

REPEAT / UNTIL

INC / DEC

DATA
RESTORE
READ

10. List of alla graphics commands

SCREEN / SELECT

SCROLL

LOAD

WAIT / VBL

DOUBLE BUFFER

CLSG

CLSG2PRINTG

INPUTG

PAPERG

INKG

LOCATEG

CONFIGFONT

PLOT

LINE

BOX / BAR

CIRCLE/CIRCLEF

FILL

TEXT

SPRITE

LOAD / POS / ADDFRAME / FRAME / HANDLE / ZOOM / ON / OFF

SAM PLAY /STOP

MUSIC PLAY

RASTER

PARTICLE

TILE

LOAD / MAP / SET / DRAW

MAP

LOAD

FONT

LOAD / MAP / PRINT / CHAR / ROTATE / ZOOM / CLEAR

11. List of alla functions

INKEY\$
STR\$
CHR\$
ASC
VAL
SIN
COS
RND
INT
HEX
INC
DEC
HIT
TILE
KEYSTATE
LEN
TRIM\$
LOWER\$
LEFT\$
RIGHT\$
MID\$
REPLACE\$
WORD\$
INSTR

12. C64 Palette Index

0 Black
1 White
2 Red
3 Cyan
4 Purple
5 Green
6 Blue
7 Yellow
8 Orange
9 Brown
10 Light red
11 Dark grey
12 Grey
13 Light green
14 Light blue